Escherichia coli Meningitis — Comprehensive Disease Characteristics Report

Disease: *Escherichia coli* meningitis **MONDO ID:** MONDO:0006753 **Category:** Infectious Disease **Report type:** Disease knowledge base entry (multi-iteration autonomous literature synthesis)

Summary

Escherichia coli meningitis is an acute bacterial infection of the meninges and cerebrospinal fluid (CSF) that is, worldwide, the **leading Gram-negative cause of neonatal bacterial meningitis**. The overwhelming majority of neonatal cases are caused by a specialized pathotype — **neonatal-meningitis *E. coli* (NMEC)** — that carries the **K1 capsular polysaccharide** and belongs predominantly to **extraintestinal pathogenic *E. coli* (ExPEC) phylogroup B2, sequence-type complex 95 (ST95)**, serotypes O18:K1, O1:K1, and O2:K1. Approximately 80% of neonatal-meningitis *E. coli* express the K1 capsule. In contemporary early-onset neonatal meningitis cohorts, *E. coli* and Group B *Streptococcus* (GBS) are co-predominant, with one 20-year cohort attributing 37.1% of blood-culture-positive cases to *E. coli* [PMID: 42661190].

The pathogenesis follows a well-defined causal chain. The K1 sialic-acid capsule confers resistance to the **alternative complement pathway**, enabling the bacterium to achieve and sustain **high-grade bacteremia**. A threshold level of bacteremia leads to **receptor-mediated transcytosis across the blood-brain barrier (BBB)** at the level of human brain microvascular endothelial cells (HBMEC), mediated by at least three independent invasin–receptor–signaling modules: **OmpA→Ecgp96(gp96)→PI3K**, **CNF1→37LRP→RhoA**, and **IbeA→vimentin/PSF→ERK–NF-κB**. Crucially, bacterial entry occurs by **transcellular transcytosis without initial disruption of endothelial tight junctions**. Downstream, invasion triggers **RIPK1/ripoptosome-dependent regulated endothelial cell death** (coordinated apoptosis,

pyroptosis, and necroptosis) together with **neutrophil (PMN) transmigration** driven by IbeA/vimentin and upregulated ICAM-1/CD44, culminating in **BBB breakdown, neuroinflammation, and neuronal injury**.

Despite modern antimicrobial therapy (third-generation cephalosporins ± aminoglycoside or meropenem for resistant strains), outcomes remain poor: **mortality of 12% (term) to 18% (preterm)**, with **long-term neurosensorial sequelae in 14–17%** and **neurodevelopmental sequelae in 10–17%** of survivors. Ampicillin resistance is high (~78%), and ESBL/ST131 clones threaten empirical regimens. **No licensed vaccine exists**; the conserved surface polysaccharide **PNAG** and **OmpA** are the leading experimental candidates, both showing proof-of-concept protection in neonatal mouse models. This report organizes 14 confirmed findings across the 15 requested disease-characteristic domains.

---

1. Disease Information

Overview. *E. coli* meningitis is an acute pyogenic (purulent) bacterial meningitis caused by the Gram-negative bacillus *Escherichia coli*. It is predominantly a **disease of the neonate** (birth–28 days, extending to young infants <90 days), although it also occurs in older adults, immunocompromised hosts, and as a nosocomial/post-neurosurgical complication (e.g., ventriculoperitoneal shunts, transsphenoidal surgery). It is the leading Gram-negative bacterial cause of neonatal meningitis worldwide [PMID: 36709579].

> "Worldwide, Escherichia coli is the leading cause of neonatal Gram-negative bacterial meningitis, but full understanding of the pathogenesis of this disease is not yet achieved." — PMID: 36709579

Key identifiers. - **MONDO:** MONDO:0006753 - **MeSH:** *Meningitis, Escherichia coli* (a subtype under *Meningitis, Bacterial* / *Central Nervous System Bacterial Infections*) - **ICD-10:** G00.8 (Other bacterial meningitis); P36/P37 codes apply in neonatal sepsis/congenital-infection contexts - **ICD-11:** 1D00 (Bacterial meningitis) with an *E. coli* causal extension code - **OMIM / Orphanet:** Not a Mendelian disorder; no OMIM phenotype number. Not assigned a distinct Orphanet rare-disease number (infectious, not genetic).

Synonyms / alternative names. *Escherichia coli* meningitis; *E. coli* meningitis; neonatal *E. coli* meningitis; *E. coli* K1 meningitis; Gram-negative enteric bacillary meningitis (when grouped); neonatal-meningitis *E. coli* (NMEC) infection of the CNS.

Information source type. This knowledge base entry is derived from **aggregated disease-level resources** — population-based surveillance, retrospective cohorts, mechanistic in vitro (HBMEC) and animal-model studies, and reviews — rather than individual EHR-level patient records.

---

2. Etiology

Primary cause (infectious). The disease is caused by infection with *Escherichia coli*, chiefly the **K1-encapsulated NMEC pathotype**. K1 strains account for ~80% of neonatal-meningitis *E. coli*.

> "Escherichia coli K1 is the leading cause of neonatal gram-negative bacterial meningitis, but the pathogenesis of E coli K1 meningitis remains unclear." — PMID: 38531686

Risk factors (host / environmental). - **Prematurity** (OR 5.84, 95% CI 2.02–16.85) and **low birth weight** (OR 8.48, 95% CI 2.60–27.69) are strong independent risk factors for death in neonatal bacterial meningitis [PMID: 29384979]. - **Neonatal age** itself — an immature immune system and immature/permeable BBB. - **Maternal genitourinary/intestinal colonization** with ExPEC (vertical transmission) driving early-onset disease; nosocomial acquisition for late-onset disease. - **Twin/multiple gestation** — shared exposure and genetic factors increase risk; twin concordance ~35% in cohort/surveillance studies [PMID: 42497927]. - **Neurosurgical devices / procedures** in older patients (VP shunts, transsphenoidal surgery, CSF leak, diabetes) [PMID: 29191537;

P 34522514

].

- **Immunocompromise** in adults (community-acquired *E. coli* meningoencephalitis/brain abscess) [PMID: 40809603].

Genetic risk factors (host). No established Mendelian human susceptibility gene is specific to *E. coli* meningitis. Host susceptibility is essentially **non-genetic**, driven by developmental immaturity. (In an animal model, loss of **Trefoil Factor 2 (Tff2)** sensitized rat pups to systemic

E. coli K1 infection [PMID: 30833331] — a model finding, not a validated human risk locus.) The critical "genetics" of this disease reside in the **pathogen genome** (virulence-gene repertoire; see Section 4).

Protective factors. Specific anti-K1/anti-capsular antibody protects against complement-resistant bacteremia (the K1 capsule evades complement only *in the absence* of specific antibody) [PMID: 6196296]. In a small twin subgroup, **pre-emptive antibiotics** for a well-appearing co-twin were associated with lower disease risk (11% vs 56%, p=0.03) [PMID: 42497927]. A probiotic *Bifidobacterium* mixture prevented neonatal *E. coli* K1 translocation in a model system [PMID: 28979247].

Gene–environment interactions. The dominant interaction is between **pathogen virulence genotype** (K1 capsule, *ompA*, *cnf1*, *ibeA*, GimA island) and the **host developmental window** (neonatal BBB permeability, immature complement/antibody repertoire). No classical human GxE locus is defined.

---

3. Phenotypes

Neonatal *E. coli* meningitis presents **nonspecifically**; classic meningeal signs are frequently absent. In a UK/Ireland population-based cohort of infants <90 days (n=263 with clinical data) [PMID: 29384979]:

Phenotype (HPO suggestion)	Frequency	Type
Fever (HP:0001945)	54%	Symptom/sign
Seizures (HP:0001250)	28%	Clinical sign
Bulging anterior fontanelle (HP:0004474 / HP:0005484)	22%	Physical sign
Coma / reduced consciousness (HP:0001259)	6%	Clinical sign
Nuchal rigidity / neck stiffness (HP:0031179)	3%	Clinical sign

> "Fever was reported in 143 (54%), seizures in 73 (28%), bulging fontanelle in 58 (22%), coma in 15 (6%) and neck stiffness in 7 (3%)." — PMID: 29384979

Other reported features include lethargy, poor feeding/failure to suck, temperature instability, apnea, irritability, and hypotonia. **Laboratory abnormalities (CSF):** neutrophilic pleocytosis, elevated CSF protein, and **low CSF glucose (hypoglycorrhachia)** (LOINC panels for CSF cell count, protein, glucose) — e.g., a severe example with CSF protein 572 mg/dL and glucose 9.5 mg/dL with abundant PMNs [PMID: 22090305].

- **Age of onset:** neonatal (early-onset ≤72 h; late-onset >72 h to ~28 days; extends to <90 days).
- **Severity:** severe, life-threatening.
- **Progression:** acute; can be fulminant with rapid deterioration.
- **Quality-of-life impact:** survivors frequently suffer hearing loss, epilepsy, cerebral palsy, hydrocephalus, and neurodevelopmental impairment, with major lifelong QoL burden. In the UK/Ireland cohort, 23% of survivors had serious CNS complications at discharge [PMID: 29384979].

Suggested HPO terms for complications: **Hydrocephalus (HP:0000238)**, **Brain abscess (HP:0031845)**, **Ventriculitis**, **Sensorineural hearing impairment (HP:0000407)**, **Cerebral palsy (HP:0100021)**, **Global developmental delay (HP:0001263)**.

---

4. Genetic/Molecular Information

Because this is an **infectious**, not inherited, disease, the relevant "genetics" are **bacterial virulence genes**, not human germline variants. There are **no causal human genes, pathogenic ACMG/AMP variants, allele frequencies, modifier genes, chromosomal abnormalities, or germline/somatic classifications** applicable to disease causation. (This section is explicitly *not applicable* in the classical Mendelian sense for a bacterial infection.)

Key bacterial virulence determinants (the effective "molecular genetics" of NMEC):

Gene / locus	Product	Role	Evidence
kpsMT / *neu* (K1 capsule)	K1 polysialic acid capsule	Complement (alternative pathway) evasion → bacteremia	[PMID: 6196296]
ompA	Outer membrane protein A	Binds Ecgp96/gp96 → PI3K → invasion	[PMID: 12860457; P 24932957]
cnf1	Cytotoxic necrotizing factor-1	Binds 37LRP → RhoA activation → BBB traversal	[PMID: 11877402; P 12860457]
ibeA (GimA island)	Invasion protein IbeA	Binds vimentin/PSF → ERK–NF-κB → invasion + PMN transmigration	[PMID: 20088823; P 22536447; P 21083634]
ibeB, *yijP*	Invasion loci	Contribute to endothelial invasion	[PMID: 11008113]
ompT	Outer membrane protease T	Adherence/invasion; virulence (APEC/NMEC)	[PMID: 28199899]
rpoE	Sigma-E stress factor	Stress resistance, invasion, pathogenicity	[PMID: 35630325]
PNAG synthesis (*pga* locus)	Poly-β-(1-6)-N-acetylglucosamine	Surface polysaccharide; virulence & vaccine target	[PMID: 36709579]

Phylogenomic background. NMEC belong predominantly to **phylogroup B2** and **ST complex 95**; among 28 US early-onset sepsis/meningitis isolates, B2 predominated (68%), of which 53% were K1-encapsulated, with phylogroup D also present (18%) [PMID: 26407251].

> "phylogroup B2 strains predominated (68%), of which more than half were K1-encapsulated (53%)" — PMID: 26407251

Epigenetic / chromosomal (host): Not applicable.

5. Environmental Information

- **Infectious agent (NCBI Taxonomy):** *Escherichia coli* (**NCBI:txid562**); pathotype NMEC; predominant capsular type **K1**; serotypes O18:K1:H7 (prototype strain RS218), O1:K1, O2:K1.
- **Environmental / lifestyle factors:** Not a toxin- or lifestyle-driven disease. Relevant "environmental" exposures are **maternal colonization**, **perinatal transmission**, **hospital/device exposure** (VP shunts, external ventricular drains, post-neurosurgical), and **contaminated water/equipment** in nosocomial neonatal settings.
- **Reservoir / transmission:** Maternal gastrointestinal and genitourinary tract; vertical transmission perinatally; horizontal/nosocomial acquisition for late-onset disease.

---

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **NMEC colonizes** the neonatal gut/mucosa and **enters the bloodstream** (vertical/perinatal acquisition). → leads to
2. **K1 sialic-acid capsule impedes activation of the alternative complement pathway** (in the absence of specific antibody), protecting the bacterium from complement-mediated killing [PMID: 6196296]. → results in
3. **High-grade, sustained bacteremia** — a threshold magnitude of bacteremia is required for CNS entry (inferred from newborn-rat models) [PMID: 6196296]. → leads to
4. **Bacterial binding to human brain microvascular endothelial cells (HBMEC)** at the BBB via multiple invasin–receptor pairs. This branches into **three independent signaling modules**:
5. **4a. OmpA → Ecgp96 (gp96, HSP90-family) → PI3K activation** (no effect on RhoA); requires GlcNAc1-4GlcNAc N-glycan epitopes on Ecgp96 [PMID: 12860457; P 24932957].
6. **4b. CNF1 → 37-kDa laminin receptor (37LRP) → RhoA activation** (no effect on PI3K); required for BBB traversal in vivo [PMID: 11877402; P 12860457].

7. **4c. IbeA → vimentin (primary receptor) + PSF (co-receptor) → CaMKII-mediated vimentin Ser82 phosphorylation, ERK phosphorylation, lipid-raft/caveolin-1 → NF-κB nuclear translocation** [PMID: 20088823; P 22536447]. → each results in
8. **Actin cytoskeletal rearrangement and receptor-mediated endocytosis → transcellular transcytosis of *E. coli* across the endothelial monolayer WITHOUT initial tight-junction disruption** [PMID: 11008113]. (Ancillary node: OmpA–Ecgp96 engagement upregulates GTP-cyclohydrolase-1 (GCH1)/biopterin/NO synthesis, promoting invasion — inferred from inhibitor studies [PMID: 23100563].) → leads to
9. **Bacterial entry into the CSF/subarachnoid space** and replication, plus **IbeA/vimentin-driven neutrophil (PMN) transendothelial migration** with upregulation of ICAM-1 and CD44 [PMID: 21083634]. → results in
10. **RIPK1-kinase-dependent assembly of the ripoptosome** in brain endothelial cells, driving **coordinated apoptosis, pyroptosis, and necroptosis** [PMID: 40774959]. → leads to
11. **BBB disruption, vasogenic edema, neuroinflammation, and neuronal injury** → clinical meningitis (fever, seizures, bulging fontanelle, coma) and, downstream, **cerebral abscess, ventriculitis, hydrocephalus, and neurodevelopmental sequelae**.

> "OmpA treatment of human BMEC resulted in phosphatidylinositol 3-kinase (PI3K) activation with no effect on RhoA, while CNF1 treatment resulted in RhoA activation with no effect on PI3K..." — PMID: 12860457
>
> "vimentin is a primary receptor required for IbeA+ E. coli K1-induced signalling and invasion of HBMECs" — PMID: 20088823
>
> "RS218 infection promotes assembly of the Ripoptosome complex. This leads to the coordinated activation of apoptosis, pyroptosis, and necroptosis." — PMID: 40774959

Upstream vs downstream

- **Upstream (bacterial-side, initiating):** K1 capsule complement evasion; bacteremia magnitude.
- **Midstream (host–pathogen interface):** invasin–receptor engagement and transcytosis (OmpA/Ecgp96, CNF1/37LRP, IbeA/vimentin).

- **Downstream (host tissue injury):** ripoptosome-driven endothelial death, PMN transmigration, cytokine/NF-κB inflammation, BBB breakdown.

Molecular pathways, processes, and ontology suggestions

- **Pathways:** PI3K–AKT (OmpA arm), RhoA/Rho-kinase (CNF1 arm), ERK–MAPK → NF-κB (IbeA arm), RIPK1/RIPK3–MLKL necroptosis and caspase-driven apoptosis/pyroptosis.
- **GO biological processes:** transcytosis (GO:0045056); regulation of Rho protein signal transduction (GO:0035023); positive regulation of NF-κB transcription factor activity (GO:0051092); necroptotic process (GO:0070266); pyroptosis (GO:0070269); leukocyte transendothelial migration (GO:0050900); inflammatory response (GO:0006954).
- **GO cellular components:** membrane raft (GO:0045121); intermediate filament / vimentin.
- **CL cell types:** brain microvascular endothelial cell (**CL:1001509**), neutrophil (**CL:0000775**), macrophage (**CL:0000235**), neuron (**CL:0000540**).

Immune involvement: NF-κB-driven pro-inflammatory cascade, PMN recruitment; the K1 capsule and OmpA engage innate evasion (complement; FcγRI/CD64 on macrophages [PMID: 26407066]). **Metabolic node:** GCH1/biopterin/NO (BH4 pathway) [PMID: 23100563]; CsiR→ilvB→FAK/PI3K under low-iron conditions promotes invasion [PMID: 38531686].

---

7. Anatomical Structures Affected

Organ / system level. - **Primary organ:** brain and meninges — **leptomeninges (UBERON:0002361)**, **subarachnoid space (UBERON:0002420)**, **cerebrospinal fluid (UBERON:0001359)**. - **Body system:** central nervous system / **meninges (UBERON:0002360)**. - **Secondary involvement:** cerebral cortex and hippocampus (microvessels show increased glycosylated Ecgp96 in infected mice [PMID: 24932957]); cerebral ventricles (**ventriculitis**), brain parenchyma (**cerebral abscess**), and systemic bloodstream (bacteremia/sepsis). Hydrocephalus and ischemo-hemorrhagic lesions are described complications [PMID: 23178136].

Tissue / cell level. - **Tissue:** cerebral microvascular endothelium (nervous-system vasculature); nervous tissue. - **Target cells: brain microvascular endothelial cells (HBMEC; CL:1001509)** — the principal cellular gateway; **neutrophils (CL:0000775)** (transmigration); **macrophages (CL:0000235)** (FcγRI/CD64 interaction); neurons and glia (injury, reactive gliosis).

Subcellular level (GO cellular component). - Plasma-membrane **membrane rafts (GO:0045121)** and **caveolae** (caveolin-1) mediate IbeA signaling; **cytoskeleton** — actin filaments, microtubules, and **intermediate filaments (vimentin)**; **nucleus** (NF-κB translocation). Ecgp96/gp96 localizes to the ER and cell surface.

Localization / lateralization. Diffuse, **bilateral** leptomeningeal involvement is typical; focal complications (abscess, infarct) may be **unilateral or asymmetric**.

---

8. Temporal Development

- **Onset age:** neonatal — **early-onset (≤72 h)** typically vertically acquired, and **late-onset (>72 h to ~28 days, up to <90 days)** often nosocomial. Also affects older adults and post-neurosurgical patients.
- **Onset pattern: acute to fulminant**; can progress within hours.
- **Stages / course:** acute bacterial meningitis → potential early complications (cerebral abscess, ventriculitis, ischemo-hemorrhagic lesions) → convalescence → long-term neurological sequelae in survivors [PMID: 23178136].
- **Progression rate:** rapid; deterioration to coma predicts death (OR 31.85) [PMID: 29384979].
- **Duration:** acute illness over days–weeks with antibiotics; **sequelae are lifelong**.
- **Critical period / window of opportunity:** early empirical antibiotics are decisive — delayed/inadequate therapy increases complications [PMID: 23178136]; delayed presentation independently predicts unfavorable outcome [PMID: 41387772].

---

9. Inheritance and Population

Epidemiology. - Neonatal *E. coli* invasive disease is **uncommon**. Population-based surveillance in the Paris area (2019–2021, ≥34 weeks gestation) recorded **early-onset neonatal sepsis incidence of 0.32 per 1000 live births (95% CI 0.26–0.38)**, of which *E. coli* meningitis is a subset; GBS was ~twice as frequent as *E. coli* in term infants [PMID: 35902218]. - Norwegian data give **neonatal bacterial meningitis incidence of 36 per 100,000 live births**, with *E. coli* responsible for 18% of childhood cases [PMID: 41273196].

> "108 EOS were recorded (annual incidence, 0.32 per 1000 live births; 95% CI 0.26 to 0.38)" — PMID: 35902218

Etiologic share (selected cohorts):

Cohort / setting	*E. coli* share	Reference
Early-onset neonatal meningitis, 20-yr cohort	37.1% of blood-culture-positive (vs GBS 30.7%)	[PMID: 42661190]
Australian tertiary, culture-proven neonatal meningitis	19% (4/21; tied with *S. aureus*)	[PMID: 41543078]
Ethiopian suspected meningitis (culture+PCR)	45.5% (25/55) — most common	[PMID: 41387772]
Norwegian childhood cases	18%	[PMID: 41273196]

Inheritance / penetrance / founder effects / carrier frequency / consanguinity: Not applicable — infectious, non-Mendelian disease.

Population demographics. - **Age:** overwhelmingly neonates/young infants; secondary peaks in older/immunocompromised adults and post-neurosurgical patients. - **Sex ratio:** male predominance is generally reported for neonatal Gram-negative sepsis/meningitis (not precisely quantified in the reviewed sources). - **Geographic distribution:** worldwide; higher burden and higher antimicrobial resistance in resource-limited settings [PMID: 41387772]. - **Pathogen lineage distribution:** phylogroup B2 / ST95 (O1/O2/O18:K1) globally; emerging multidrug-resistant ST131 clones.

---

10. Diagnostics

Gold standard — CSF analysis plus culture. Definitive diagnosis requires **lumbar puncture** with CSF showing **neutrophilic pleocytosis, elevated protein, and low glucose (hypoglycorrhachia)**, plus a **positive CSF and/or blood culture** growing Gram-negative rods [PMID: 22090305].

Test	Finding in *E. coli* meningitis	Notes
CSF cell count/differential (LOINC)	Neutrophilic pleocytosis	Nonspecific bacterial pattern
CSF protein (LOINC)	Elevated (e.g., 572 mg/dL)	[PMID: 22090305]
CSF glucose (LOINC)	Low (hypoglycorrhachia; e.g., 9.5 mg/dL)	[PMID: 22090305]
CSF/blood culture	Gram-negative bacilli; *E. coli*	Gold standard; enables susceptibility testing
CSF Gram stain	Gram-negative rods	Rapid presumptive
Blood culture	Positive in high-grade bacteremia	Supports diagnosis

Molecular / omics adjuncts. - **Multiplex PCR panels** (e.g., MeltArray CNS targeting 85 meningitis pathogens) and syndromic panels detect *E. coli* directly from CSF, outperforming culture especially after antibiotics [PMID: 42623488; P 41387772].

- **Targeted metagenomic next-generation sequencing (mNGS)** of CSF shows **moderate agreement (~59% positive agreement) with conventional diagnostics** and aids detection in culture-negative cases, but high unexpected-detection rates limit standalone use [PMID: 42636981].

> "Targeted mNGS shows moderate agreement with conventional diagnostics and may aid pathogen detection in selected patients with negative routine testing." — PMID: 42636981

Imaging. Cranial ultrasound (neonate; ventricular dilatation), CT/MRI to detect abscess, ventriculitis, infarction, hydrocephalus.

Genetic testing / newborn screening / host omics diagnostics: Not applicable (no host genetic cause). "Screening" here refers to microbiologic detection, not carrier/germline testing.

Differential diagnosis. GBS meningitis (co-predominant; distinguished by Gram stain/culture), *Listeria monocytogenes*, *Klebsiella pneumoniae* and other Enterobacterales, *Citrobacter*, *S. aureus*, viral/enteroviral meningitis, and non-infectious causes.

11. Outcome / Prognosis

Mortality. Outcome is poor: **12% mortality in term and 18% in preterm** infants [PMID: 23178136]. Population cohorts report ~9–19% case fatality [PMID: 29384979;].

P 41543078

> "Outcome of early and late onset E. coli neonatal meningitis is poor with 12% (term infant) to 18% (premature infant) mortality rates." — PMID: 23178136

Morbidity / sequelae. Long-term **neurosensorial sequelae 14–17%** and **neurodevelopmental sequelae 10–17%** [PMID: 23178136]. Early complications: cerebral abscess, ventriculitis, ischemo-hemorrhagic lesions, hydrocephalus. In one cohort, 23% of survivors had serious CNS complications at discharge [PMID: 29384979].

> "Long term sequelae, particularly neurosensorial [14-17%] and neurodevelopmental [10-17%] are frequent." — PMID: 23178136

Prognostic factors. Independent predictors of **death**: **prematurity (OR 5.84)**, **low birth weight (OR 8.48)**, and **coma at presentation (OR 31.85)** [PMID: 29384979]. Predictors of **CNS complications**: seizures (OR 7.06), temperature instability (OR 2.99), high CSF protein (OR 2.62) [PMID: 29384979]. Additional adverse factors: hemodynamic failure, apnea, hypoglycorrhachia, abnormal EEG, and delayed/inadequate antibiotics [PMID: 23178136]. In broader cohorts, low Glasgow Coma Scale, comorbidity, and delayed presentation independently predict unfavorable outcome [PMID: 41387772].

12. Treatment

Pharmacotherapy (mainstay). - **Empirical:** third-generation cephalosporins — **cefotaxime (NCIT:C376)** or **ceftriaxone (NCIT:C769)** — often combined with an **aminoglycoside (gentamicin/amikacin)** or **ampicillin (NCIT:C263)** in neonatal regimens. - **Resistant strains: meropenem (NCIT:C1352)** for ESBL/AmpC producers; carbapenems for cephalosporin

resistance. - **Rationale for coverage choice: ampicillin resistance is high (~78%)** among NMEC, so ampicillin monotherapy is inadequate empirically [PMID: 26407251]. Contemporary ESBL-producing and ST131/CTX-M-15 clones further threaten empirical regimens [PMID: 26407251;

41820864

].

> "the rate of ampicillin resistance was high (78%)" — PMID: 26407251

Adjunctive therapy. - **Corticosteroids (dexamethasone)** reduce mortality and neurological sequelae in acute bacterial meningitis overall (chiefly pneumococcal); however, **benefit in neonatal Gram-negative meningitis is not established**, and guidelines generally exclude neonates [PMID: 41675892;

42601017

]. - **Supportive care:** anticonvulsants for seizures, management of raised intracranial pressure/hydrocephalus (may require CSF diversion/shunt), hemodynamic and respiratory support.

Experimental / host-directed therapeutics (model-organism evidence). - **Mesenchymal stem cells (MSCs):** intraventricular human umbilical-cord-blood-derived MSC transplantation reduced CSF bacterial growth and mortality and attenuated brain injury in newborn-rat *E. coli* meningitis [PMID: 30188499].

> "Intraventricular transplantation of hUCB-MSCs significantly improved survival and attenuated the brain injury via anti-inflammatory and antibacterial effects in experimental neonatal E. coli meningitis." — PMID: 30188499

- **MSC-derived extracellular vesicles (MSC-EVs)** similarly attenuated brain injury [PMID: 35888118].
- **Mechanism-targeted candidates (preclinical):** RIPK1 inhibitors (to block ripoptosome-driven endothelial death) [PMID: 40774959]; GCH1/biopterin inhibitor DAHP (prevented invasion and meningitis in newborn mice) [PMID: 23100563]; anti-OmpA/Ecgp96 interface blockade [PMID: 24932957].

Pharmacogenomics / gene / cell / RNA / targeted / immuno-therapy (host): Not applicable as disease-specific approved therapies.

13. Prevention

Primary prevention. - **Maternal screening and intrapartum management** of colonization; hygiene and infection-control in neonatal units to prevent nosocomial acquisition. - **Pre-emptive antibiotics** for well-appearing co-twins of an affected neonate reduced disease risk in a small subgroup (11% vs 56%, p=0.03) [PMID: 42497927]. - **Probiotics:** a *Bifidobacterium* mixture prevented neonatal *E. coli* K1 translocation/infection in a model [PMID: 28979247].

Immunization (experimental — no licensed vaccine). - **No vaccine is currently available** against bacterial neonatal meningitis [PMID: 36709579]. - **PNAG (Poly-β-(1-6)-N-Acetyl-Glucosamine):** a TnSeq-identified conserved surface polysaccharide; **both passive and active immunization prevented/treated *E. coli* K1 meningitis in neonatal mice**, with opsonophagocytic killing and reduced BBB crossing; PNAG is also expressed by GBS, offering potential cross-protection [PMID: 36709579].

> "We showed that both passive and active immunization successfully prevented and/or treated meningitis caused by E. coli K1 in neonatal mice." — PMID: 36709579

- **OmpA / Ecgp96 GlcNAc1-4GlcNAc interface:** a preventive target — disrupting this interaction blocks invasion [PMID: 24932957].

Secondary/tertiary prevention. Early lumbar puncture and prompt empirical antibiotics (secondary); management of complications (hydrocephalus, seizures, sequelae) and rehabilitation (tertiary). **Counseling / genetic screening:** Not applicable.

14. Other Species / Natural Disease

- **Taxonomy of pathogen:** *Escherichia coli* (**NCBI:txid562**).
- **Zoonotic / cross-host lineage:** Neuropathogenic *E. coli* belong to the **zoonotic-capable ExPEC/APEC phylogroup B2, ST complex 95** lineage. **Avian pathogenic *E. coli* (APEC)** and human ExPEC of serotypes O1, O2, O18 cluster in B2/ST95 with **no or minimal host specificity** [PMID: 25397580].

> "Avian pathogenic E. coli and human extraintestinal pathogenic E. coli serotypes O1, O2 and O18 strains isolated from different hosts are generally located in phylogroup B2 and ST complex 95, and they share similar genetic characteristics and pathogenicity, with no or minimal host specificity." — PMID: 25397580

- **Molecular identity across hosts:** Representative O18 APEC and NMEC strains have **identical OmpA sequences**, survive human serum equally, use the same receptors (FcγRI-α/CD64 on macrophages; Ecgp96 on endothelium), and induce meningitis in newborn mice [PMID: 26407066].

> "The amino acid sequences of the outer membrane protein A (OmpA), an important virulence factor in the pathogenesis of meningitis, are identical within these representative APEC and NMEC strains." — PMID: 26407066

- **Natural / veterinary disease:** APEC causes avian colibacillosis (poultry); the shared virulence repertoire underscores a **One Health / zoonotic reservoir** concern (poultry as a potential ExPEC reservoir).
- **Comparative biology:** the OmpA–Ecgp96, CNF1–37LRP, and IbeA–vimentin mechanisms are conserved between avian and human strains, supporting evolutionary conservation of the meningitis mechanism.

---

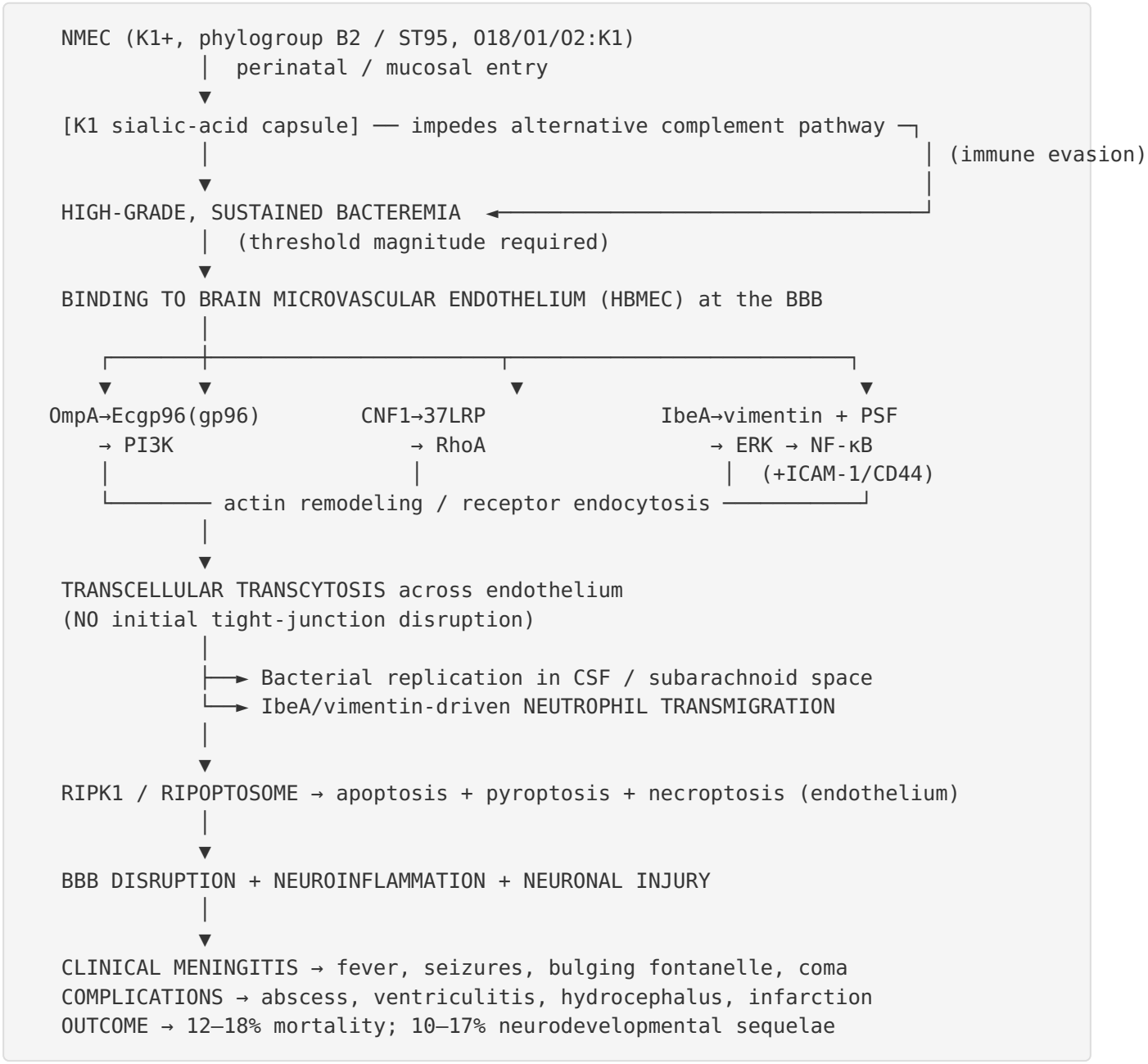
15. Model Organisms

- **Mammalian in vivo models: Newborn/neonatal rat** and **newborn mouse** models of hematogenous *E. coli* K1 meningitis are the workhorses — used to demonstrate CNF1/RhoA-dependent BBB traversal [PMID: 11877402], DAHP/biopterin-dependent meningitis prevention [PMID: 23100563], PNAG vaccine efficacy [PMID: 36709579], and MSC therapy [PMID: 30188499]. A **newborn-rat ventriculitis model without concomitant bacteremia** (intraventricular K1-negative *E. coli*) isolates CNS injury from systemic sepsis [PMID: 31846163].
- **In vitro / cellular models: Human brain microvascular endothelial cells (HBMEC)** are the standard BBB model for invasion/transcytosis and signaling studies [PMID: 12860457;

]; P 20088823 mouse **bEnd.3** BMEC used for APEC studies [PMID: 28199899]; CHO-Lec1 glycosylation-mutant cells for receptor-epitope mapping [PMID: 24932957].

- **Phenotype recapitulation:** Neonatal rodent models faithfully reproduce bacteremia → BBB crossing → CSF infection → brain injury/inflammation and neurodevelopmental impairment, and respond to interventions predictably.
- **Limitations:** Rodent BBB and immune ontogeny differ from human; K1-negative ventriculitis models bypass the natural hematogenous route; in vitro HBMEC monolayers cannot fully capture in vivo neuroinflammatory dynamics.
- **Resources:** MGI (mouse), RGD (rat); strain resources for prototype NMEC (RS218, O18:K1:H7).

Mechanistic Model / Interpretation



The central interpretive insight is that **immune evasion (capsule) and endothelial invasion (three redundant invasin pathways) are the two rate-limiting, upstream nodes**, while **RIPK1-driven regulated cell death and neutrophil influx are the downstream effectors of tissue damage**. The redundancy of invasion pathways (OmpA, CNF1, IbeA each independent, with additive inhibitor effects and greater attenuation in double mutants) explains why single-

target anti-virulence strategies may be insufficient and why **conserved capsular/surface antigens (PNAG, K1) and shared receptors (Ecgp96) are attractive vaccine/therapeutic targets**.

Evidence Base

PMID	Contribution	Evidence type
36709579	*E. coli* leading Gram-negative neonatal meningitis cause; PNAG vaccine POC; no licensed vaccine	Review + animal model
38531686	K1 leading cause; CsiR/low-iron invasion pathway	In vitro/mechanistic
42661190	Modern etiologic breakdown (*E. coli* 37.1%)	Human cohort (20-yr)
6196296	K1 capsule impedes alternative complement pathway	In vitro + newborn-rat
12860457	OmpA→PI3K vs CNF1→RhoA independent invasion pathways	In vitro (HBMEC)
11877402	CNF1/RhoA required for in vivo BBB traversal	Animal model
20088823	Vimentin is primary IbeA receptor for invasion	In vitro (HBMEC)
22536447	Vimentin+PSF drive NF-κB nuclear translocation	In vitro
21083634	IbeA/vimentin → PMN transmigration, ICAM-1/CD44	In vitro
11008113	Transcytosis route; ompA/ibeA/ibeB/yijP gene set	Review/mechanistic
40774959	RIPK1/ripoptosome-driven endothelial death & BBB disruption	In vitro/model
24932957	OmpA–Ecgp96 GlcNAc epitope; preventive target	In vitro + mouse
23100563	GCH1/biopterin/NO node; DAHP prevents meningitis	In vitro + newborn mice
29384979	Phenotype frequencies; death/CNS-complication predictors	Population cohort (UK/Ireland)
23178136	Mortality 12–18%; sequelae 10–17%	Review
26407251	Phylogroup B2 68%/K1 53%; ampicillin resistance 78%; ST131/CTX-M-15	US isolates
35902218	EOS incidence 0.32/1000 live births	Population surveillance (Paris)
41273196	Neonatal meningitis 36/100,000; *E. coli* 18%	National (Norway)
25397580	B2/ST95 APEC–ExPEC shared lineage, minimal host specificity	Comparative genomics

PMID	Contribution	Evidence type
26407066	Identical OmpA, shared receptors APEC/NMEC	Comparative in vitro/ model
30188499	MSC therapy improves survival/brain injury	Newborn-rat model
35888118	MSC-EVs attenuate brain injury	Model
41675892	Corticosteroid adjunct benefit (mainly pneumococcal; excludes neonates)	Narrative review
42636981	Targeted mNGS diagnostic role & limits	Diagnostic study
42623488	Multiplex PCR (MeltArray CNS) for CSF pathogens	Diagnostic study
22090305	Severe CSF profile example (protein 572, glucose 9.5)	Case report
42497927	Twin concordance; pre-emptive antibiotics	Systematic review

Limitations and Knowledge Gaps

1. **Not a genetic disease.** Sections geared to Mendelian genetics (causal genes, ACMG variants, inheritance, penetrance, founder effects, genetic testing, newborn screening) are **not applicable**; the "genetics" reside in the pathogen. Host susceptibility loci for *E. coli* meningitis are essentially undefined in humans.
2. **Mechanism largely from models.** Much of the invasion/injury mechanism (OmpA/CNF1/IbeA pathways, RIPK1 ripoptosome, biopterin node) derives from **in vitro HBMEC and neonatal-rodent models**; direct human validation is limited.
3. **Epidemiologic heterogeneity.** *E. coli* share of neonatal meningitis varies widely by setting (18–46%); resource-limited settings show higher resistance and worse outcomes, complicating global generalization.
4. **No head-to-head neonatal Gram-negative corticosteroid trial.** Adjunctive steroid benefit is extrapolated from predominantly pneumococcal data and is **not established for neonatal Gram-negative meningitis**.
5. **Resistance trajectory.** ESBL/ST131/CTX-M-15 emergence data are from limited isolate collections; the true trajectory threatening empirical cephalosporin therapy needs continued surveillance.

6. **Vaccine translation gap.** PNAG and OmpA are promising but remain **preclinical**; no human efficacy data exist.
7. **Sex-ratio and QoL quantification** were not precisely resolved from the reviewed literature.

Proposed Follow-up Experiments / Actions

1. **Pathogen genomic surveillance:** Systematically genotype NMEC isolates (phylogroup, ST, capsule, ESBL/carbapenemase genes) across regions to track ST131/ST95 dynamics and inform empirical regimens.
2. **RIPK1-targeted host-directed therapy:** Test RIPK1 kinase inhibitors (e.g., necrostatin-class) as adjuncts in neonatal-rodent *E. coli* meningitis to quantify reductions in BBB disruption and neurodevelopmental sequelae [building on

   P 40774959

   ].
3. **Multivalent conserved-antigen vaccine:** Advance PNAG (+ OmpA/Ecgp96-interface) conjugate vaccines toward safety/immunogenicity studies, leveraging cross-protection against GBS PNAG [PMID: 36709579].
4. **Combination anti-invasin strategy:** Given pathway redundancy, test simultaneous blockade of ≥2 invasion arms (PI3K + Rho-kinase + IbeA/vimentin) in vivo for additive protection [PMID: 12860457].
5. **Neonatal Gram-negative steroid/adjunct trial:** Design a dedicated trial or observational study of dexamethasone or alternative immunomodulation specifically in neonatal Gram-negative meningitis.
6. **Diagnostic validation:** Prospectively validate multiplex-PCR and targeted mNGS CSF panels against culture in neonatal cohorts to define sensitivity for culture-negative (pre-treated) cases [PMID: 42623488;

   P 42636981

   ].
7. **One Health investigation:** Quantify poultry/APEC B2-ST95 reservoirs and potential zoonotic transmission routes for neuropathogenic *E. coli* [PMID: 25397580].

Report compiled from 14 confirmed findings and 56 reviewed papers across a 5-iteration autonomous investigation. Evidence types are labeled (human clinical, animal model, in vitro, computational/genomic) throughout. All quoted text is verbatim from cited PubMed abstracts.
